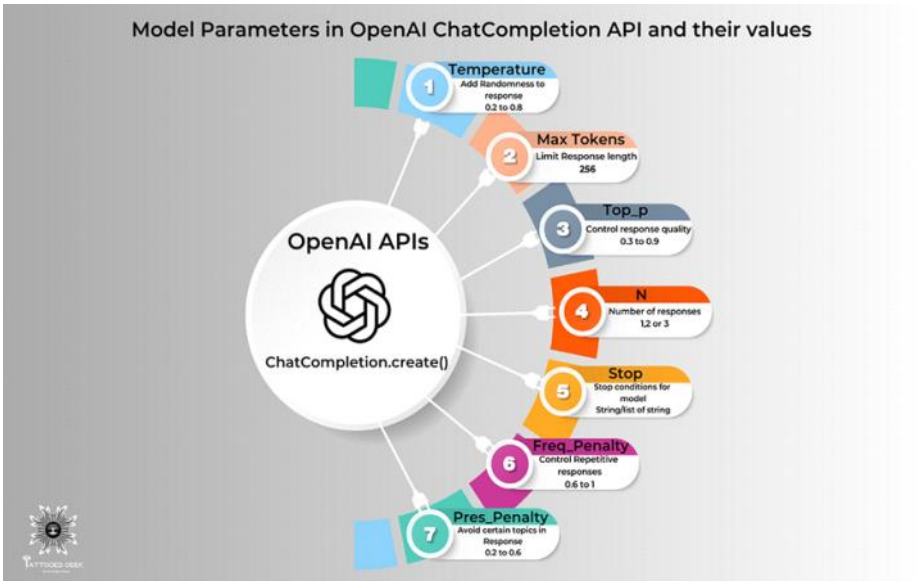
Model Parameters in OpenAI API

Let’s break down the ChatCompletion.create() Function and find the sweet-spot for the parameter values together.
High-Level overview of model Parameters and their Values



Introduction

In this blog post, we will delve into the inner workings of the client.chat.completions.create() function in the OpenAI API.This function plays a crucial role in generating most coherent responses from the ChatGPT model.By understanding the various parameters it accepts and fine-tuning them, you can control the behavior and quality of the model’s responses.

1. Understanding the client.chat.completions.create() Function

This function allows us to generate responses from the ChatGPT model by providing a series of messages as input.
In the client.chat.completions.create() the function, the strings **“system”**, **“user”**, and **“assistant”** are used to define the role of each message within the conversation. These roles help provide context and structure to the model’s understanding of the conversation flow.
The function call looks like this:

```
from openai import OpenAI
client = OpenAI()
completion = client.chat.completions.create(model="gpt-3.5-turbo",
messages=[{"role": "system", "content": "You are a helpful assistant."},{"role": "user", "content": "Hello!"}])print(completion.choices[0].message)
```

Here, we pass the model parameter specifying the model (e.g., “**get-3.5-turbo**”). The messages parameter is an array of message objects, each having a “**role**” (“**system”, “user”, or “assistant**”) and “**content**” (the actual text of the message).

1. “system”: The “system” role typically provides high-level instructions or context-setting messages. It helps guide the behavior of the assistant throughout the conversation. For example, a system message could be “You are a helpful assistant.”
2. “user”: The “user” role represents the messages or queries from the user or end-user interacting with the model. These messages contain the user’s input or questions. For instance, a user message could be “Tell me a fun fact.”
3. “assistant”: The “assistant” role represents the responses generated by the ChatGPT model. It includes the model’s generated replies or information provided to the user based on their queries. For example, an assistant message could be, “Sure, here is an interesting fact:”

By assigning specific roles to each message, you establish a conversational context and guide the model’s understanding of the user’s input and the assistant’s responses. This allows for a more interactive and coherent conversation between the user and the AI model.

High-Level overview of model Parameters and their Values | Designed in Canva by

2. Temperature: Adding Randomness to the Responses

The temperature parameter influences the randomness of the generated responses. A higher value, such as 0.8, makes the answers more **diverse**, while a lower value, like 0.2, makes them more **focused** and **deterministic**.

A value between 0.2 and 0.8 can be effective. Lower values (e.g., 0.2) produce more focused and deterministic responses, while higher values (e.g., 0.8) allow for more randomness.

Code Sample:

```
response = client.chat.completions.create (model="gpt-3.5-turbo",messages=[{"role": "user", "content": "Tell me a joke."},{"role": "assistant", "content": "Why don't scientists trust atoms?"},{"role": "user", "content": "I don't know, why?"},{"role": "assistant", "content": "Because they make up everything!"}],temperature=0.8)
```

Output:

```
{…“choices”: [{“message”: {“content”: “Why did the chicken cross the road? To get to the other side!”,“role”: “assistant”}}],…}
```

3. Max Tokens: Limiting the Response Length

The max_tokens parameter allows you to limit the length of the generated response. Setting an appropriate value allows you to control the response length and ensure it fits the desired context.

Tokens: In NLP, a token refers to a unit of text the model reads. Tokens can be as short as one character or as long as one word, depending on the language and the specific model used. Tokens are common sequences of characters found in the text.

Rule of thumb: 1 token is approximately 4 characters or 0.75 words for English text.

GPT models excel at understanding token relationships and generating the next token in a sequence.

The model processes text by reading and generating tokens, and the number of tokens in an API call affects the cost and response time. Hence we need to set the max_tokens parameter

and put a limit on the response length.
It's essential to remember that tokens are not equivalent to words, as they can also represent punctuation marks, spaces, or special characters.

Code Sample:

```
response = client.chat.completions.create (model="gpt-3.5-turbo",messages=[{"role": "user", "content": "Translate the following English text to French: 'Hello, how are you?'"},{"role": "assistant", "content": "Sure, here is the translation: 'Bonjour, comment ça va ?'"}],max_tokens=20)
```

Output:

```
{…"choices": [{"message": {"content": "Bonjour, comment ça","role": "assistant"}}],…}
```

4. Top P (Nucleus Sampling): Controlling Response Quality

The top_p parameter, also known as nucleus sampling or "penalty" in the API, controls the diversity and quality of the responses. It limits the cumulative probability of the most likely tokens. Higher values like **0.9** allow more tokens, leading to **diverse** responses, while lower values like **0.2** provide more **focused** and **constrained** answers..

Code Sample:

```
response = client.chat.completions.create (model="gpt-3.5-turbo",messages=[{"role": "user", "content": "Who was the first man on the moon?"},{"role": "assistant", "content": "The first man on the moon was Neil Armstrong."},{"role": "user", "content": "Tell me more about him."}],top_p=0.5)
```

Output:

```
{…"choices": [{"message": {"content": "Neil Armstrong was an astronaut who became famous for being the first person to walk on the moon. He was a part of the Apollo 11 mission in 1969.","role": "assistant"}}],…}
```

5. n: Generating Multiple Responses

The n parameter allows you to generate multiple alternative completions for a given conversation. By increasing the value of n, you can explore different response variations. Experiment with different values (e.g., 1, 2, or 3) to specify the number of alternative completions to generate. Increasing this value can offer more diverse response options.

Code Sample:

```
response = client.chat.completions.create (model="gpt-3.5-turbo",messages=[{"role": "system", "content": "You are a helpful assistant."},{"role": "user", "content": "Who is the best soccer player of all time?"}],n=3)
```

Output:

```
{'choices': [{'message': {'role': 'assistant','content': 'There are many great soccer players, but some of the most frequently mentioned ones include Pelé, Diego Maradona, and Lionel Messi.'},'finish_reason': 'stop','index': 0},{'message': {'role': 'assistant','content': 'When it comes to soccer, opinions may vary, but some popular choices for the best player of all time include Pelé, Diego Maradona, and Cristiano Ronaldo.'},'finish_reason': 'stop','index': 1},{'message': {'role': 'assistant','content': 'The best soccer player of all time is subjective and can vary depending on personal preferences. However, some widely recognized legends in the sport include Pelé, Diego Maradona, and Lionel Messi.'},'finish_reason': 'stop','index': 2}]}
```

6. Stop: Customizing stop Conditions

The stop parameter allows you to specify a custom condition for the completion. You can provide a string or a list of strings that, when encountered in the generated response, will cause the model to stop generating further tokens. Here's an example
Providing a list of stop words can help prevent the model from generating responses containing those specific words. Include relevant stop words based on your application.

Code Sample:

```
response = openai.ChatCompletion.create(model="gpt-3.5-turbo",messages=[{"role": "system", "content": "You are a helpful assistant."},{"role": "user", "content": "Translate the following English text to French: 'Hello, how are you?'"}],stop="
```

Translate:")

Output:

{'id': 'chatcmpl-6p9XYPYSTTRi0xEviKjjilqrWU5Ve','object': 'chat.completion','created': 1677649420,'model': 'gpt-3.5-turbo','usage': {'prompt_tokens': 56, 'completion_tokens': 62, 'total_tokens': 118},'choices': [{'message': {'role': 'assistant','content': 'Salut, comment ça va ?'},'finish_reason': 'stop','index': 0}]}

7. Frequency Penalty: Controlling Repetitive Responses

The frequency_penalty parameter allows you to control the model's tendency to generate repetitive responses. Higher values, like **1.0**, **encourage the model to explore more diverse and novel responses**, while lower values, such as **0.2,** make the **model more likely to repeat information.**

Increasing this value (e.g., 0.6) encourages the model to avoid repeating the same words/phrases and can lead to more varied responses.

Code Sample:

```
response = client.chat.completions.create (model="gpt-3.5-turbo",messages=[{"role": "user", "content": "What are some tourist attractions in Paris?"},{"role": "assistant", "content": "Some popular attractions in Paris are the Eiffel Tower, Louvre Museum, and Notre-Dame Cathedral."},{"role": "user", "content": "Tell me more about the Louvre Museum."}],frequency_penalty=0.6)
```

Output:

{…"choices": [{"message": {"content": "The Louvre Museum is one of the world's largest and most visited museums. It is located in the heart of Paris and houses a vast collection of art and historical artifacts.","role": "assistant"}}],…}

8. Presence Penalty: Controlling Avoidance of Certain Topics

The presence_penalty parameter allows you to influence the model's avoidance of specific topics in its responses. **Higher values, such as 1.0, make the model more likely to avoid mentioning particular** topics provided in the user messages, while lower values, like **0.2**, make the **model less concerned about preventing those topics**.

Increasing this value (**e.g., 0.6**) encourages the model to include more relevant details from the provided context and can enhance the specificity of responses.

Code Sample:

```
response client.chat.completions.create (model="gpt-3.5-turbo",messages=[{"role": "user", "content": "Tell me about your favorite book."},{"role": "assistant", "content": "I enjoy reading science fiction novels."},{"role": "user", "content": "What are your thoughts on 'Dune' by Frank Herbert?"}],presence_penalty=0.8)
```

Output:

{…"choices": [{"message": {"content": "I'm sorry, but I am not familiar with 'Dune' by Frank Herbert.","role": "assistant"}}],…}

9. Implementing Parameters in Python Code

In this section, we provide a Python code snippet that combines the aforementioned parameters, allowing you to experiment and observe their effects on the generated responses.

Code Sample:

```
response = openai.ChatCompletion.create(model="gpt-3.5-turbo",messages=[{"role": "user", "content": "Tell me a fun fact."},{"role": "assistant", "content": "Sure, here is an interesting fact:"}],temperature=0.6,max_tokens=30,top_p=0.8,frequency_penalty=0.6,presence_penalty=0.8)
```

The expected output would be the response generated by the chatbot, which would be a completion of the conversation based on the provided context and the behavior of the model with the given parameters.